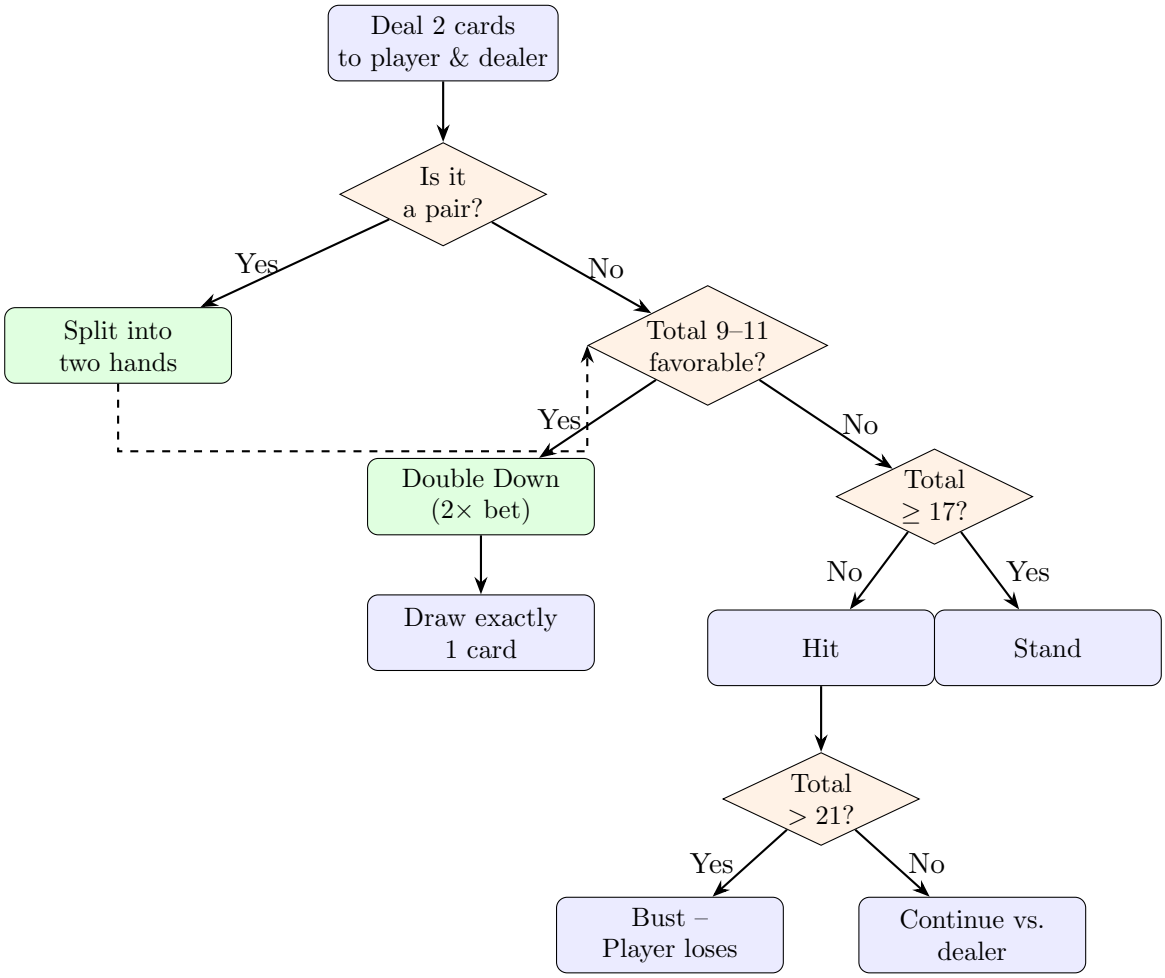
Blackjack Probability Overview

Alvin Wang

6/09/26

1 Decision Flow (Hit / Stand / Double / Split)

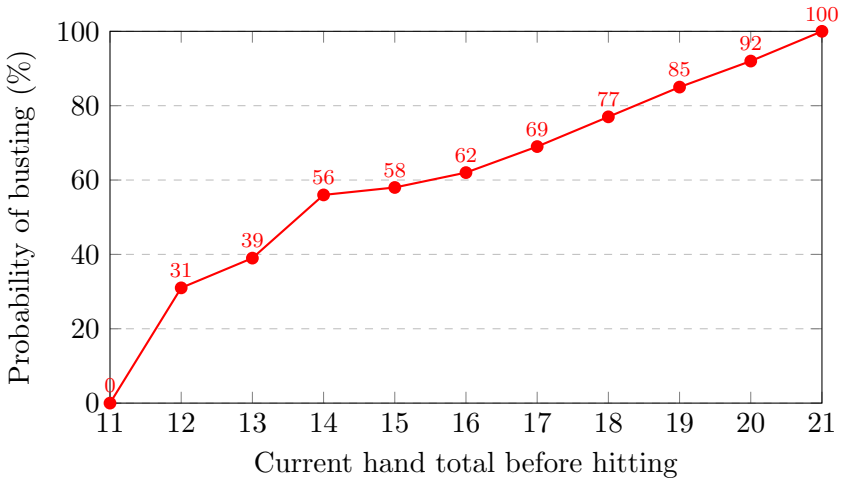
A simplified diagram of the choices a player faces after seeing their initial two cards.



Note: Each split hand is then played independently through the same double/hit/stand logic; the dashed arrow indicates that a split hand re-enters the flow.

2 Bust Probability by Hand Total

The chance of busting (going over 21) if the player chooses to hit, as a function of current hand total, assuming a standard deck.



3 Common Outcome Probabilities

Approximate probabilities for a few standard blackjack scenarios (single deck, no other strategy adjustments).

Event	Description	Probability
Natural Blackjack	Ace + 10-value card on first two cards	4.83%
Dealer Bust	Dealer busts overall (all upcards combined)	28.36%
Player Bust on 16	Player hits a 16 and busts	62%
Push (Tie)	Player and dealer end with same total	8.48%
Player Wins (overall)	Average win rate with basic strategy	42.22%

4 Estimated Win Probability by Starting Hand

Approximate probability the player wins the hand outright, given the starting two cards and assuming basic-strategy play afterward.

Starting Hand	Description	Win Probability
Natural Blackjack	Ace + 10-value card	≈92%
Hard 20	e.g. 10 + 10	≈70%
Hard 11	Typically doubled down	≈66%
Soft 18	e.g. Ace + 7	≈52%
Pair of Aces	Split, one card each	≈52%
Hard 17	e.g. 10 + 7	≈30%
Pair of 8s	Split, one card each	≈44%
Hard 16	e.g. 10 + 6	≈24%

5 Basic Strategy Chart

Legend: H = Hit, S = Stand, D = Double Down (Hit if doubling is not allowed), P = Split. Columns are the dealer's face-up card. This is a commonly cited single-deck chart (dealer stands on soft 17); exact optimal play can shift slightly with rule variations.

Hard Totals

Player	2	3	4	5	6	7	8	9	10	A
5–8	H	H	H	H	H	H	H	H	H	H
9	H	D	D	D	D	H	H	H	H	H
10	D	D	D	D	D	D	D	D	H	H
11	D	D	D	D	D	D	D	D	D	H
12	H	H	S	S	S	H	H	H	H	H
13	S	S	S	S	S	H	H	H	H	H
14	S	S	S	S	S	H	H	H	H	H
15	S	S	S	S	S	H	H	H	H	H
16	S	S	S	S	S	H	H	H	H	H
17+	S	S	S	S	S	S	S	S	S	S

Soft Totals (hand includes an Ace counted as 11)

Player	2	3	4	5	6	7	8	9	10	A
A,2 (13)	H	H	H	D	D	H	H	H	H	H
A,3 (14)	H	H	H	D	D	H	H	H	H	H
A,4 (15)	H	H	D	D	D	H	H	H	H	H
A,5 (16)	H	H	D	D	D	H	H	H	H	H
A,6 (17)	H	D	D	D	D	H	H	H	H	H
A,7 (18)	S	D	D	D	D	S	S	H	H	H
A,8+ (19+)	S	S	S	S	S	S	S	S	S	S

Pairs

Player	2	3	4	5	6	7	8	9	10	A
2,2	P	P	P	P	P	P	H	H	H	H
3,3	P	P	P	P	P	P	H	H	H	H
4,4	H	H	H	D	D	H	H	H	H	H
5,5	D	D	D	D	D	D	D	D	H	H
6,6	P	P	P	P	P	H	H	H	H	H
7,7	P	P	P	P	P	P	H	H	H	H
8,8	P	P	P	P	P	P	P	P	P	P
9,9	P	P	P	P	P	S	P	P	S	S
10,10	S	S	S	S	S	S	S	S	S	S
A,A	P	P	P	P	P	P	P	P	P	P

Note: All figures above are illustrative approximations commonly cited for basic-strategy play and will vary slightly with number of decks and house rules.